



Northern Beaches Secondary College
Manly Campus

2019 HIGHER SCHOOL CERTIFICATE EXAMINATION
Trial Examination

Mathematics

**General
Instructions**

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- NESA approved calculators may be used
- A reference sheet is provided
- In Questions 11 – 16, show relevant mathematical reasoning and/ or calculations

**Total marks:
100**

Section I – 10 marks (pages 2-6)

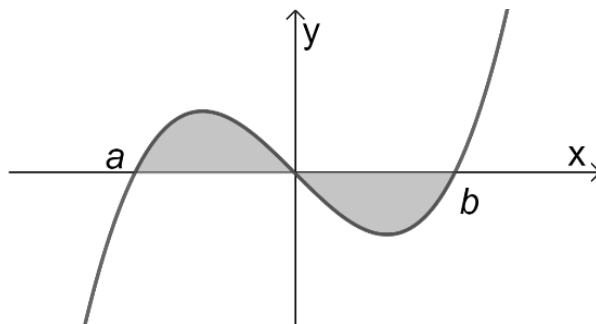
- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 7-17)

- Attempt Questions 11 – 16
- Allow about 2 hours and 45 minutes for this section

Section I - 10 marks**Attempt Questions 1 – 10****Allow about 15 minutes for this section**Use the multiple-choice answer sheet for Questions 1-10

- 1 The graph of a function $y = f(x)$, where $f(-x) = -f(x)$ is shown below. The graph intersects the x -axis at $x = a$, $x = 0$ and $x = b$.



Which one of the below gives the area of the shaded region?

- A. $\int_a^b f(x) \, dx$
- B. $2 \int_a^{a+b} f(x) \, dx$
- C. $2 \int_0^{-a} f(x) \, dx$
- D. $-2 \int_b^0 f(x) \, dx$
- 2 The gradient of the normal to the curve $f(x) = 3x^3 - 4x + 2$ at the point $(-1, 3)$ is:
- A. 5
- B. -5
- C. $-\frac{1}{5}$
- D. $-\frac{1}{3}$

- 3 It is known that for a particular quadratic function, $\alpha + \beta = -\frac{5}{3}$ and $\alpha\beta = \frac{7}{3}$. The quadratic function could be:

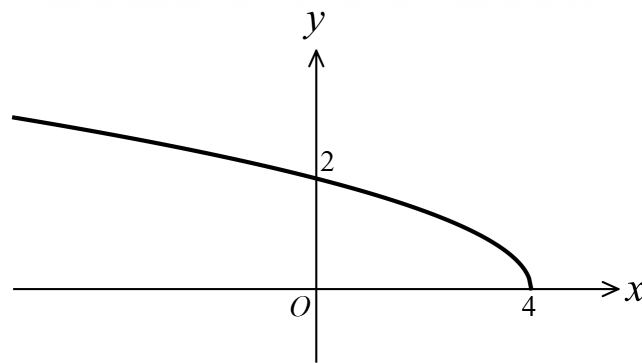
A. $f(x) = 3x^2 - 5x + 7$

B. $f(x) = 3x^2 + 5x + 7$

C. $f(x) = 3x^2 + 5x - 7$

D. $f(x) = 5x^2 - 7x + 3$

- 4 The graph of the curve $y = \sqrt{4-x}$ is shown below.



For what value(s) of x is the curve not differentiable?

- A. $x < 4$
B. $x = 4$
C. $0 < x < 4$
D. $x = 0$
- 5 Consider the points A (1, -2) and B (3, 6). What is the equation of the perpendicular bisector of AB?

A. $y - 2 = -\frac{1}{4}(x - 2)$

B. $y - 2 = 4(x - 2)$

C. $y - 4 = -1(x - 1)$

D. $y + 2 = -\frac{1}{4}(x - 1)$

6 It is given that $\ln a = 2 \ln b + \ln c - \ln d$. Which of the following statements is true?

A. $\ln a = \ln \left(2b + \frac{c}{d} \right)$

B. $\ln a = \frac{cb^2}{d}$

C. $a = \frac{b^2c}{d}$

D. $\ln a = \frac{\ln(b^2 + c)}{\ln d}$

7 If $\int_0^2 f(x) dx = 6$ what is the value of $\int_0^2 [x - 2f(x)] dx$?

A. -10

B. -5

C. 8

D. 12

8 Which of the following possible k values will there be two distinct rational solutions for the equation.

$$2x^2 + 20x + k = 0$$

A. $k = 52$

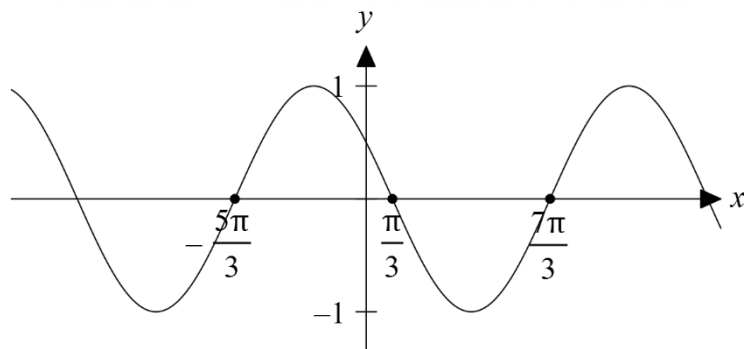
B. $k = 50$

C. $k = 48$

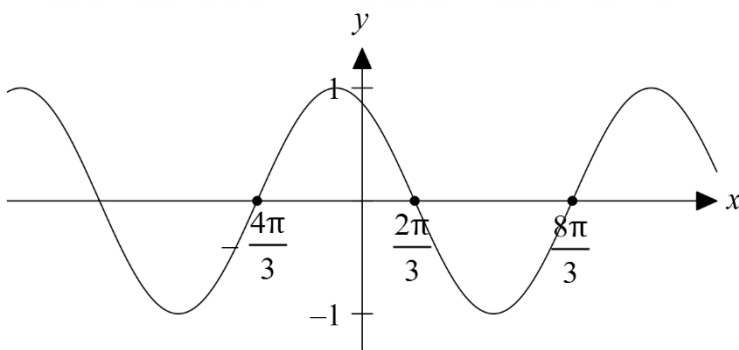
D. $k = 46$

9 Which diagram shows the graph $y = \cos\left(\frac{x}{2} + \frac{\pi}{3}\right)$?

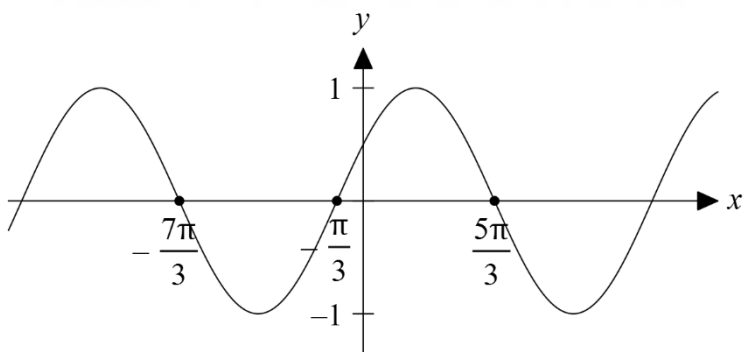
A.



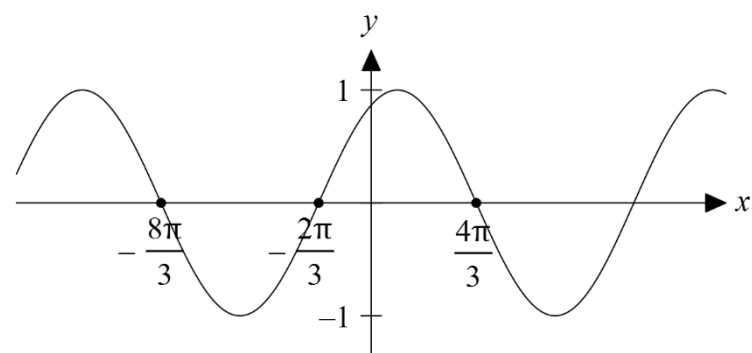
B.



C.

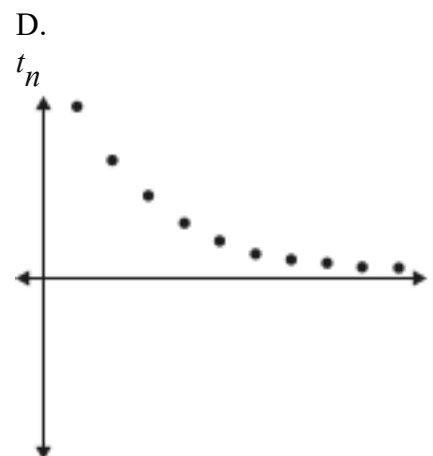
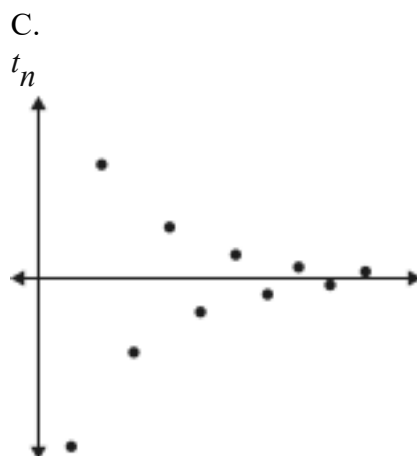
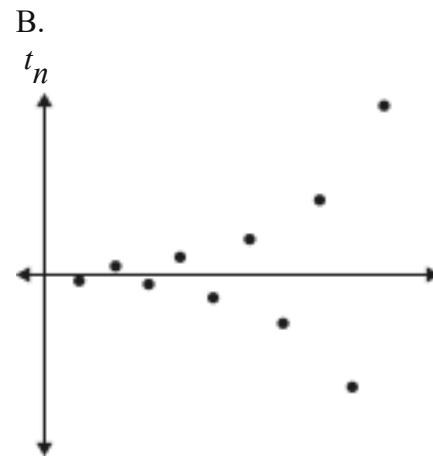
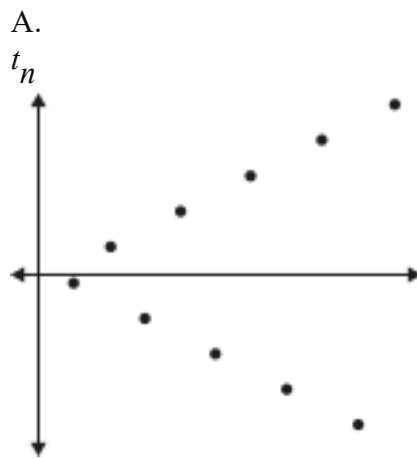


D.



- 10** The first term of a geometric sequence is a , where $a < 0$. The common ratio of this sequence, r , is such that $r < -1$.

Which one of the following graphs best shows the first 10 terms of this sequence?



End of Multiple Choice

Section II

90 marks

Attempt Questions 11 – 16

Allow about 2 hours and 45 minutes for this section

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

In Questions 11 – 16, your response should include relevant mathematical reasoning and/ or calculations.

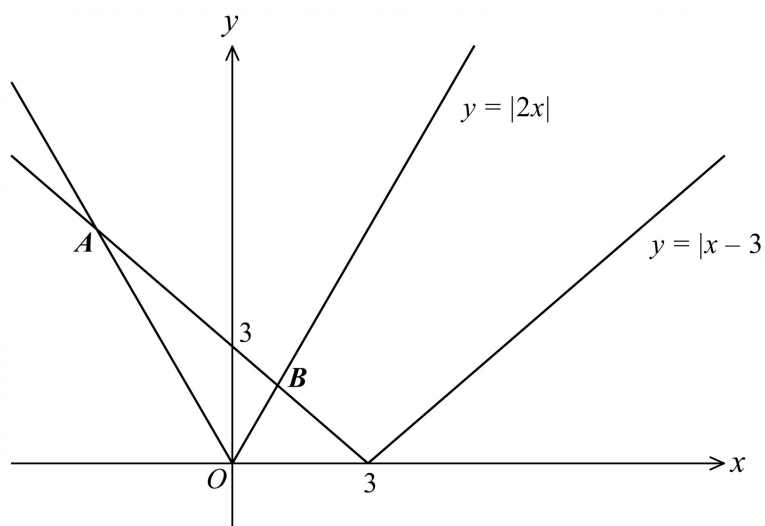
Question 11 (15 marks) Use the Question 11 Writing Booklet	Marks
(a) Evaluate $\int_1^5 (3x - 7) dx$	2
(b) Solve $ 2x - 5 < 7$	2
(c) Find the limiting sum of the geometric series, $2 - \frac{1}{3} + \frac{1}{18} - \frac{1}{108} + \dots$	2
(d) Differentiate with respect to x	
i. $\frac{x^2}{e^{2x}}$	2
ii. $\cos^3(5x + 3)$.	2
(e) Determine $\int \frac{3dx}{(3 - 2x)^3}$	2
(f) Find the coordinates of the focus of the parabola $y = -\frac{1}{8}x^2 + x - 1$.	3

End of Question 11

- (a) Find the point on the graph of $f(x) = x^2 - 5x + 4$ where the tangent has a gradient of -3. 2

- (b) Given $y = x\sqrt{1-x^2}$, show is $\frac{dy}{dx} = \frac{1-2x^2}{\sqrt{1-x^2}}$ 3

- (c) The diagram shows the graphs of $y = |2x|$ and $y = |x - 3|$ which intersect at points A and B .



- (i) Find the x - coordinate of points A and B . 2
- (ii) Hence or otherwise, solve $|x - 3| \leq |2x|$ 1
- (d) Given $y = \ln(\cos x)$
- (i) Differentiate $\ln(\cos x)$ with respect to x . 1
- (ii) Hence evaluate $\int_0^{\frac{\pi}{4}} \tan x \, dx$ 2

Question 12 continues on next page

Question 12 continued**Marks**

- (e) The temperature T , in degrees Celsius, of an oven is given by the equation

$$T = 120 + 80\sin\left(\frac{\pi}{3}t\right) \text{ where } t \text{ is measured in hours.}$$

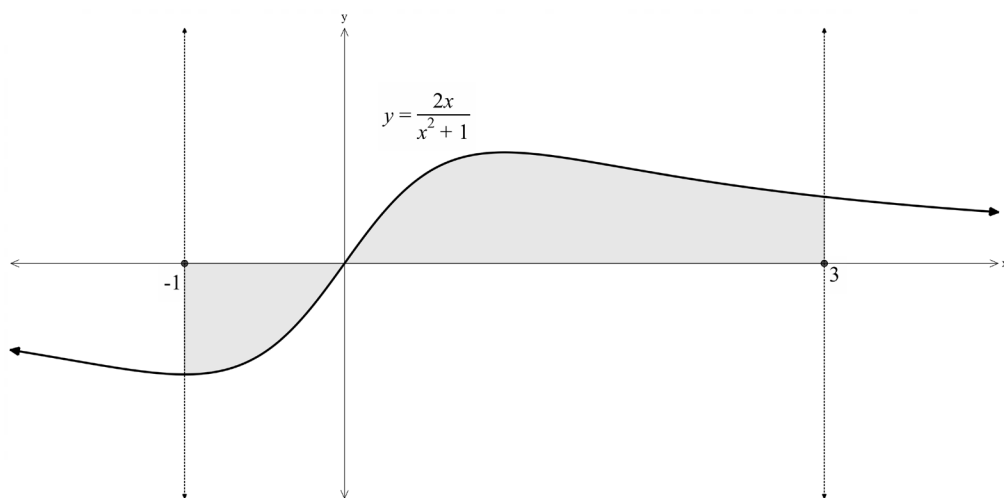
- (i) What is the range of temperature in the oven? **1**
- (ii) What is the rate of change of the temperature when the oven first reaches two thirds of its initial temperature? Leave your answer in exact form. **3**

End of Question 12

(a) Prove the following $\frac{1 - \tan^2 x}{1 + \tan^2 x} = 1 - 2\sin^2 x$ 3

(b) The diagram shows the graph of $f(x) = \frac{2x}{x^2 + 1}$. The shaded area is

enclosed by the curve $f(x) = \frac{2x}{x^2 + 1}$, the x -axis, $x = -1$ and $x = 3$



Use Simpson's Rule and the 5 function values to approximate the shaded area.

Give the answer correct to 1 decimal place.

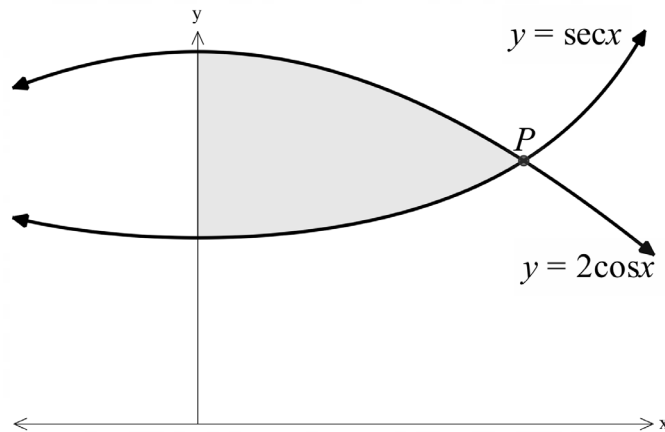
3

(c) Gallium-67 is a radioactive element that decays over time.

The mass of gallium-67 after t hours is given by $M(t) = Ae^{kt}$.

- i. Show that $M(t)$ satisfies $\frac{dM}{dt} = kM$ 1
- ii. Given that the half-life of gallium-67 is 80 hours, find the value of k to 3 significant figures. 2
- iii. How many hours will it take for a mass of gallium-67 to reach 10% of its original mass? 2

- (d) The diagram below shows the shaded area bounded by the curves $y = \sec x$, $y = 2\cos x$ and the y -axis.



- i. Show that the x -coordinate of P is $x = \frac{\pi}{4}$ **1**
- ii. Given that $\cos^2 x = \frac{1}{2}(\cos 2x + 1)$, determine the volume of the solid of revolution produced by rotating the shaded area around the x -axis. **3**
(Give your answer in exact form)

End of Question 13

(a) Consider the function $f(x) = x(x - 2)^4$

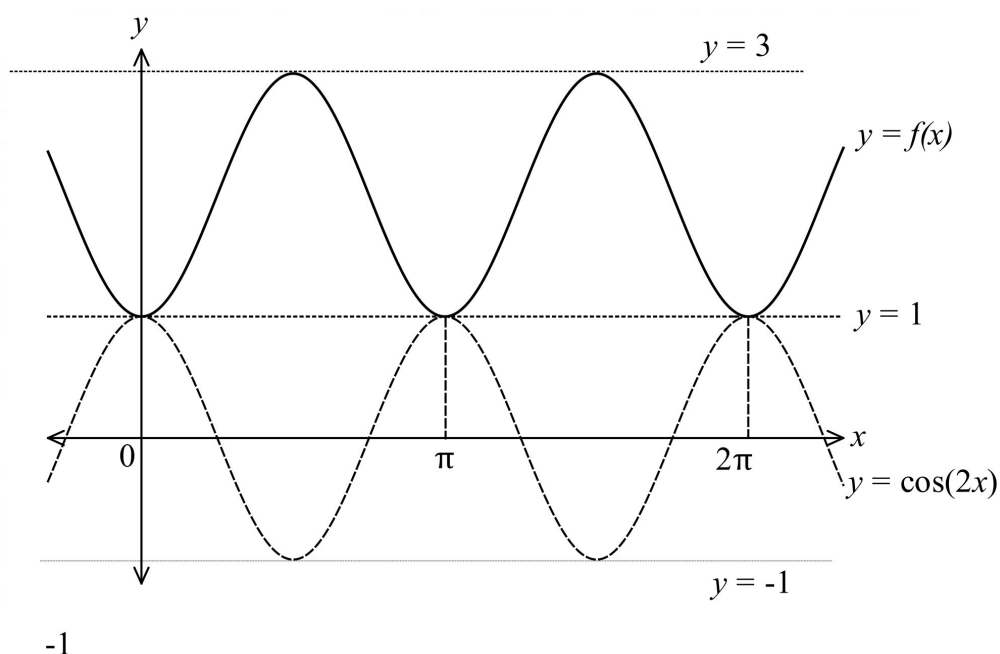
(i) Show that $f'(x) = (x - 2)^3(5x - 2)$. 1

(ii) Find the coordinates of the stationary points of $y = f(x)$ and determine their nature. 2

(iii) Sketch the graph of $y = f(x)$, showing the intercepts and stationary points. 2

(b) The diagram shows the graphs of $y = \cos(2x)$ and $y = f(x)$ from $x = 0$ to $x = 2\pi$.

The graph of $y = f(x)$ is a reflection of $y = \cos 2x$ along the line $y = 1$.



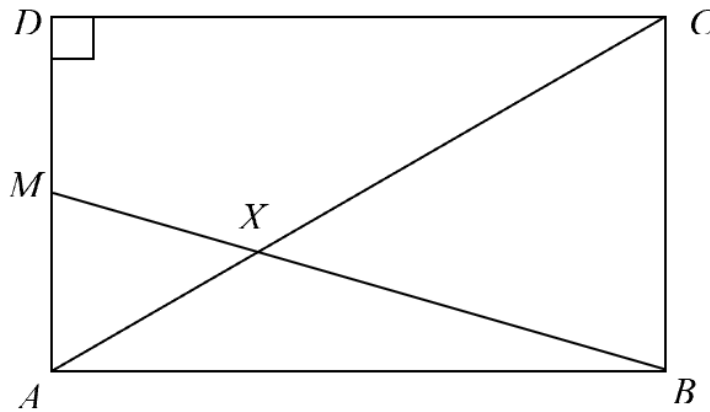
(i) Determine the equation for the graph of $y = f(x)$ 2

(ii) Find the exact area between the curves $y = \cos(2x)$ and $y = f(x)$ from $x = 0$ and $x = 2\pi$ 3

Question 14 continues on the next page

Question 14 continued**Marks**

- (c) In the diagram, $ABCD$ is a rectangle and $AB = 2AD$. The point M is the midpoint of AD . The line BM meets AC at X .



NOT TO SCALE

- | | |
|--|----------|
| (i) Prove that the triangle AXM and CXB are similar. | 2 |
| (ii) Hence show that $3CX = 2AC$. | 1 |
| (iii) Show that $9(CX)^2 = 5(AB)^2$ | 2 |

End of Question 14

(a) Show $\int_0^{\sqrt{e}} \frac{x^2}{x^3 + e} dx = \frac{\ln(\sqrt{e} + 1)}{3}$ 3

- (b) Jacqueline is a Year 12 student who plans to travel after completing the HSC exams.

On the 1st January 2019, Jacqueline has \$4000 in her savings account however she aims to have a **total** of \$7000 in her account by the 1st December 2019.

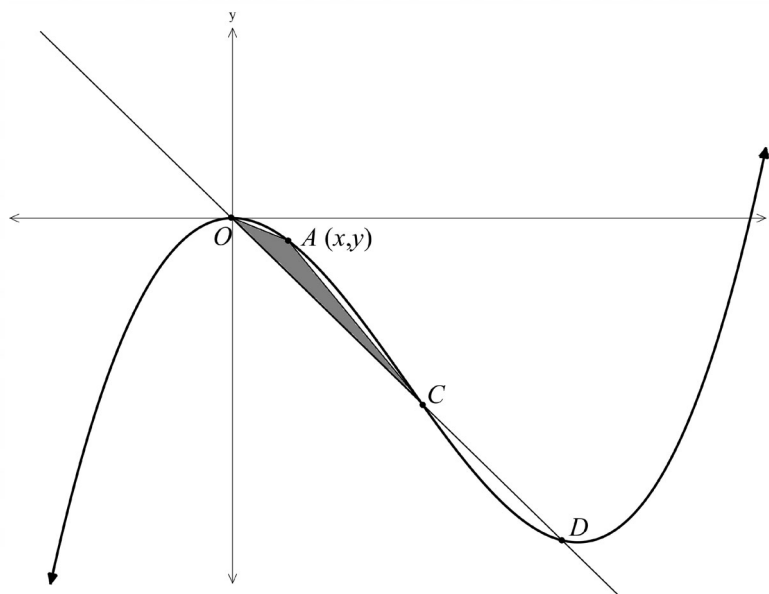
She will make monthly deposits of \$ M at the **end** of every month with the first deposit to be made on the 31st January 2019 and the last deposit to be made on the 30th November 2019.

The bank pays 6% p.a. interest compounded monthly.

- (i) How much will the \$4000 in Jacqueline's account accumulate to by 1st December 2019? Give the answer correct to the nearest cent. 1
- (ii) Calculate Jacqueline's monthly deposit, \$ M , so that she has a **total** of \$7000 in her savings account by 1st December 2019. Give the answer correct to the nearest cent. 3

Question 15 continues on the next page

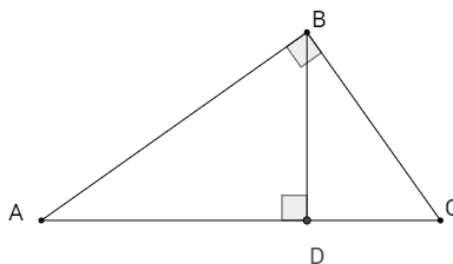
- (c) In the diagram the curve $y = 4x^3 - 9x^2$ and the line $y = -5x$ intersect at points origin O , C and D . Point A lies on the curve $y = 4x^3 - 9x^2$ between points O and C .



- (i) Determine the coordinates of C . 2
- (ii) Show that the perpendicular distance between point A and the line segment OC is $\frac{1}{\sqrt{26}}(4x^3 - 9x^2 + 5x)$. 2
- (iii) Show that the area of the triangle AOC is $2x^3 - \frac{9}{2}x^2 + \frac{5}{2}x$. 1
- (iv) The point A is chosen so that the area of the triangle AOC is a maximum.
Find the maximum possible area correct to two decimal places. 3

End of Question 15

- (a) In the diagram, triangle ABC is given such that $AB \perp BC$ and $BD \perp AC$.

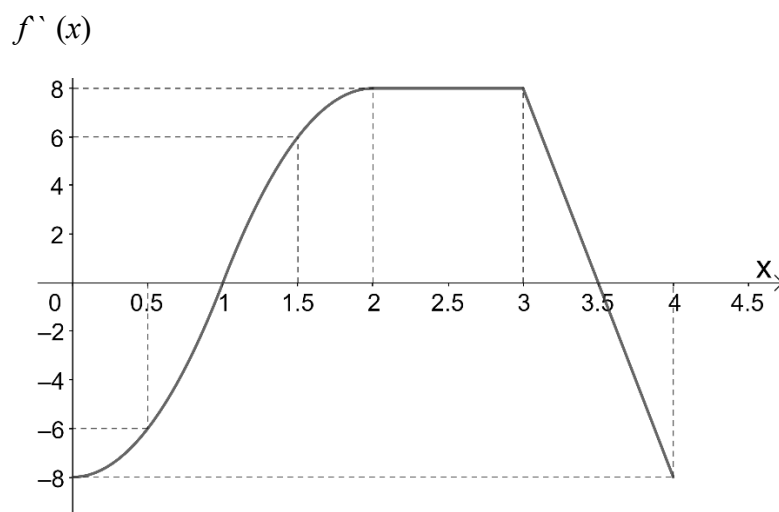


- i. By stating an appropriate triangle similarity, show that $AB^2 = AC \cdot AD$. 1
(Note: You do **NOT** have to prove similarity)
- ii. Hence, or otherwise, show that $AB^2 + BD^2 = AD(AD + 2CD)$. 2

- (b) Let $f(x)$ be a function defined for $0 \leq x \leq 4$ such that $f(0) = 0$.

The diagram shows the graph of its derivative, $y = f'(x)$.

It is known that $\int_0^2 f'(x) dx = 0$.



- (i) For which values of x is $f(x)$ decreasing? 1
- (ii) Find $f(4)$. 1
- (iii) What is the maximum value of $f(x)$? 1
- (iv) Draw a graph of $y = f(x)$ for $0 \leq x \leq 4$. 2

Question 16 continues on the next page

(c) Beatrice takes out a home loan of \$520,000.

The loan is charged reducible interest of 8.4% per annum, calculated monthly.

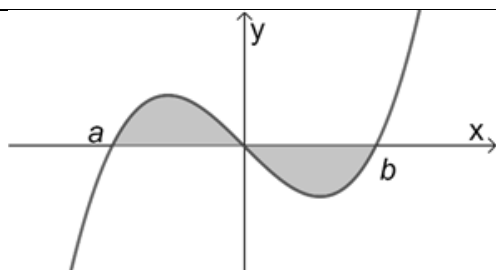
The loan is to be repaid in equal monthly repayments of \$ M over 15 years.

Let A_n be the amount owing after the n th repayment.

- | | | |
|-------|---|----------|
| (i) | Derive an expression for A_3 , the amount owing after 3 months. | 2 |
| (ii) | Show that the monthly repayment is approximately \$5090.21. | 2 |
| (iii) | Immediately after her 24 th payment, Beatrice makes a one-off payment of \$20,000. If the interest rate and monthly repayment remain unchanged, after how many more months will Beatrice pay off the loan? | 3 |

End of Examination

Multiple Choice



As $f(x)$ is odd then

1. $b = -a$

2. Let $\int_a^0 f(x) dx = k$

$$\therefore \int_0^b f(x) dx = -k$$

3. Area $= k + |-k| = 2k$

A. $\int_a^b f(x) dx = k + (-k) = 0$

B.
$$\begin{aligned} & 2 \int_a^{a+b} f(x) dx \\ &= 2 \int_a^{a-a} f(x) dx \\ &= 2 \int_a^0 f(x) dx \\ &= 2k \text{ -- CORRECT} \end{aligned}$$

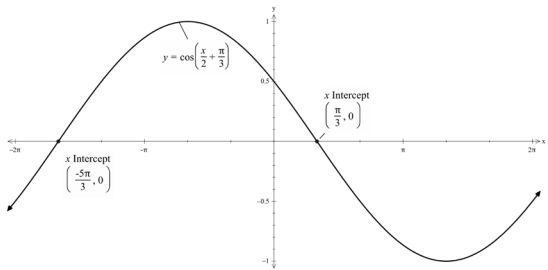
C.
$$\begin{aligned} & -2 \int_b^0 f(x) dx \\ &= -2 \times - \int_0^b f(x) dx \\ &= -2 \times -1 \times -k \\ &= -2k \end{aligned}$$

D.
$$\begin{aligned} & 2 \int_0^{-a} f(x) dx \\ &= 2 \int_0^b f(x) dx \\ &= 2 \times -k \\ &= -2k \end{aligned}$$

Q1

B

Q2	$f(x) = 3x^3 - 4x + 2$ $f'(x) = 9x^2 - 4$ $f'(-1) = 9 - 4$ $= 5$ Therefore gradient of normal is $-\frac{1}{5}$	C
Q3	$f(x) = 3x^2 + 5x + 7$	B
Q4	From the graph the tangent is vertical where $x = 4$	B
Q5	$m = -\frac{3-1}{6+2} = -\frac{1}{4}$ $\text{midpoint} = \frac{1+3}{2}, \frac{-2+6}{2}$ $= (2, 2)$ $y - 2 = -\frac{1}{4}(x - 2)$	A
Q6	$\ln a = \ln b^2 + \ln c - \ln d$ $= \ln \left[\frac{b^2 c}{d} \right]$ $\therefore a = \frac{b^2 c}{d}$	C
Q7	$\int_0^2 x - 2f(x) \, dx$ $= \int_0^2 x \, dx - 2 \int_0^2 f(x) \, dx$ $= \left[\frac{x^2}{2} \right]_0^2 - 2(6)$ $= \left(\frac{2^2}{2} - \frac{0^2}{2} \right) - 2(6)$ $= -10$	A

Q8	$2x^2 + 20x + k = 0$ Distinct means $\Delta > 0$ Rational means Δ is a perfect square. $\Delta = b^2 - 4ac$ $= 20^2 - 4(2)(k)$ $= 400 - 8k$ If $k = 48$, then $\Delta = 16$, which is both positive and a perfect square.	C
Q9	$y = \cos\left(\frac{x}{2} + \frac{\pi}{3}\right)$ $0 \leq \frac{x}{2} + \frac{\pi}{3} \leq 2\pi$ $-\frac{\pi}{3} \leq \frac{x}{2} \leq \frac{5\pi}{3}$ $-\frac{2\pi}{3} < x \leq \frac{10\pi}{3}$ Shift left of $\frac{2\pi}{3}$ 	A
Q10	Let r be an integer $\therefore$ $r \Rightarrow r^2 \Rightarrow r^3 \Rightarrow r^2 \Rightarrow r^5$ $-2 \Rightarrow 4 \Rightarrow -8 \Rightarrow 16 \Rightarrow -32$ Therefore Option B	B

Question 11

11a	$\int_1^5 (3x - 7) \, dx$ $= \left[\frac{3x^2}{2} - 7x \right]_1^5$ $= \frac{75}{2} - 35 - \left(\frac{3}{2} - 7 \right)$ $= 8$	2 marks correct answer 1 mark correct integration Note: this does not equate to an area so no units
b	$ 2x - 5 < 7$ $-7 < 2x - 5 < 7$ $-2 < 2x < 12$ $-1 < x < 6$	2 marks correct answer 1 mark correct boundary values Note: stating separate inequalities requires an 'and' not an 'or' $6 > x > -1$ is never acceptable!
c	$2 - \frac{1}{3} + \frac{1}{18} - \frac{1}{108} + \dots$ $S_{\infty} = \frac{2}{1 - -\frac{1}{6}}$ $= \frac{12}{7}$	2 marks correct solution 1 mark correct ratio or cfe - carry forward error
d i	$\frac{x^2}{e^{2x}}$ $u = x^2$ $u' = 2x$ $v = e^{-2x}$ $v' = -2e^{-2x}$ $y' = 2xe^{-2x} + -2e^{-2x}x^2$ $= \frac{2x(1-x)}{e^{2x}}$	2 marks correct answer 1 mark correct u' and v'

ii	$\cos^3(5x + 3) = (\cos(5x + 3))^3$ note $\cos^3(5x + 3) \neq \cos(5x + 3)^3$ $f(x) = \cos(5x + 3)$ $f'(x) = -5\sin(5x + 3)$ $y' = 3 \times -5\sin(5x + 3)\cos^2(5x + 3)$ $= -15\sin(5x + 3)\cos^2(5x + 3)$	2 marks correct solution 1 mark recognition of chain rule and attempt to derive
e	$\int \frac{3dx}{(3-2x)^3} \quad \text{this is not in } \frac{f'(x)}{f(x)} \text{ form}$ $= 3 \int (3-2x)^{-3} dx$ $= \frac{3}{-2 \times -2} (3-2x)^{-2} + c$ $= \frac{3}{4} (3-2x)^{-2} + c$	2 marks correct answer 1 mark an error in integration
f	$y = -\frac{1}{8}x^2 + x - 1$ $y + 1 = -\frac{1}{8}x^2 + x$ $-8(y + 1) = x^2 - 8x$ $-8(y + 1) + 16 = x^2 - 8x + 16$ $-8y + 8 = (x - 4)^2$ $(x - 4)^2 = -8(y - 1)$ $\hat{V}(4, 1) \quad a = 2, \quad a > 0$ concave down $\therefore S$ is (4,-1)	3 marks correct answer 2 marks correct vertex and focal length or cfe from vertex 1 mark correct vertex form

Question 12

a	$f'(x) = 2x - 5$ $2x - 5 = -3$ $2x = 2$ $x = 1$ $f(1) = 1 - 5 + 4$ $= 0$ point is (1,0)	2 marks for correct working and solution, also carry error if everything is correct. 1 mark for correct gradient.
b	$\frac{dy}{dx} = \sqrt{1-x^2} + x \left(-\frac{x}{\sqrt{1-x^2}} \right)$ $= \sqrt{1-x^2} - \frac{x^2}{\sqrt{1-x^2}}$ $= \frac{1-x^2-x^2}{\sqrt{1-x^2}}$ $= \frac{1-2x^2}{\sqrt{1-x^2}}$	3 marks for correct working and solution. 2 marks for 1 error. 1 mark for 2 errors.
c (i)	Point A $-2x = 3 - x$ $x = -3$ Point B $2x = 3 - x$ $x = 1$	1 mark for each x coordinate
c (ii)	From the graph $x \geq 1$ $x \leq -3$	1 mark
d (i)	$y = \ln(\cos x)$ $\frac{dy}{dx} = \frac{-\sin x}{\cos x}$ $= -\tan x$	1 mark

d (ii)	$-\int_0^{\frac{\pi}{4}} \tan x \, dx$ $= -\ln \cos \frac{\pi}{4} + \ln \cos 0$ $= -\ln \frac{1}{\sqrt{2}} + \ln 1$ $= \ln \sqrt{2}$	2 marks for working and solution 1 mark for correct integral substitution.
e (i)	Range is 40°C to 200°C	1 mark
e (ii)	$\frac{2}{3} \times 120 = 80$ $80 = 120 + 80 \sin\left(\frac{\pi}{3} t\right)$ $-0.5 = \sin\left(\frac{\pi}{3} t\right)$ $\frac{\pi}{3} t = \frac{7\pi}{6}$ $t = 3.5$ $\frac{dT}{dt} = \frac{80\pi}{3} \cos\left(\frac{\pi}{3} t\right)$ when $t = 3.5$ $\frac{dT}{dt} = \frac{80\pi}{3} \cos\left(\frac{\pi}{3} \times 3.5\right)$ $= \frac{80\pi}{3} \times -\frac{\sqrt{3}}{2}$ $= -\frac{40\sqrt{3}\pi}{3}$	3 marks for correct working and solution. 2 marks for correct derivative and $t = 3.5$ 1 mark for either correct derivative or $t = 3.5$

Question 13

a	$\frac{1 - \tan^2 x}{1 + \tan^2 x} = 1 - 2\sin^2 x$ $\begin{aligned} LHS &= \frac{1 - \tan^2 x}{\sec^2 x} \\ &= \frac{1}{\sec^2 x} - \frac{\tan^2 x}{\sec^2 x} \\ &= \cos^2 x - \left(\frac{\sin^2 x}{\cos^2 x} \times \left(\frac{1}{\sec^2 x} \right) \right) \\ &= \cos^2 x - \left(\frac{\sin^2 x}{\cos^2 x} \times (\cos^2 x) \right) \\ &= \cos^2 x - \sin^2 x \\ &= (1 - \sin^2 x) - \sin^2 x \\ &= 1 - 2\sin^2 x \end{aligned}$	3 marks- correct solution 2 marks- significant correct progress with one error ONLY 1 mark- correct use of one fundamental identity that leads to simplification
b	$f(x) = \frac{2x}{1 + x^2}$ $\begin{aligned} A &= \frac{1}{3} \left\{ \frac{6}{10} + 1 + 4 \left(\frac{4}{5} + 0 \right) \right\} \\ &= \frac{1}{3} \left\{ \frac{8}{5} + \frac{16}{5} + 2 \right\} \\ &= \frac{1}{3} \left(\frac{34}{5} \right) \\ &= \frac{34}{15} \\ &= 2.27 \end{aligned}$	3 marks- correct solution 2 marks- significant correct progress with one error ONLY without QS 1 mark- correct h and function values

ci	$M(t) = Ae^{kt}$ $\frac{dM}{dt} = kAe^{kt}$ $= kM \text{ since } M = Ae^{kt}$	1 mark-correct answer
cii	$M(80) = Ae^{k(80)}$ $\frac{A}{2} = Ae^{80k}$ $\frac{1}{2} = e^{80k}$ $80k = \ln\left(\frac{1}{2}\right)$ $k = \frac{\ln\frac{1}{2}}{80}$ $= -0.008664\dots$ -0.00866	2 marks-correct answer 1 mark-ONLY one error in correct progress to value of k without QS 1 mark-ONLY error in sig figure rounding incorrect
ciii	$0.1A = Ae^{-0.0139t}$ $0.1 = e^{-0.008669t}$ $-0.00866t = \frac{\ln 0.1}{\ln e}$ $t = \frac{\ln(0.1)}{-0.00866}$ $= 265.75\dots$ $= 266\text{hrs}$	2 marks-correct answer 1 mark-ONLY one error in correct progress to value of t without QS
di	$2\cos x = \sec x$ $2\cos x = \frac{1}{\cos x}$ $2\cos^2 x = 1$ $\cos^2 x = \frac{1}{2}$ $\cos x = \pm \frac{1}{\sqrt{2}}$ $x = \frac{\pi}{4} \text{ (since } x > 0 \text{ and first point } \int)$	1 mark-correct answer

dii	$ \begin{aligned} V &= \pi \int_0^{\frac{\pi}{4}} (2\cos x)^2 - (\sec x)^2 dx \\ &= \pi \int_0^{\frac{\pi}{4}} 4\cos^2 x - \sec^2 x dx \\ &= \pi \left\{ \left[2 \int_0^{\frac{\pi}{4}} \cos 2x + 1 \right] - \int_0^{\frac{\pi}{4}} \sec^2 x \right\} \\ &= \pi \left\{ 2 \left[\frac{1}{2} \sin 2x + x \right]_0^{\frac{\pi}{4}} - [\tan x]_0^{\frac{\pi}{4}} \right\} \\ &= \pi \left\{ [\sin 2x + 2x]_0^{\frac{\pi}{4}} - [\tan x]_0^{\frac{\pi}{4}} \right\} \\ &= \pi \left\{ \left[\sin 2 \left(\frac{\pi}{4} \right) + \frac{2\pi}{4} \right] - (\sin 0 + 0) \right\} - \left[\tan \frac{\pi}{4} - \tan 0 \right] \\ &= \pi \left\{ \left[1 + \frac{\pi}{2} \right] - 0 - [(1 - 0)] \right\} \\ &= \pi \left\{ 1 + \frac{\pi}{2} - 1 \right\} \\ &= \pi \left\{ \frac{\pi}{2} \right\} \\ &= \frac{\pi^2}{2} \end{aligned} $	3 marks- correct solution 2 marks- significant correct progress with one error ONLY 1 mark- correct answer for sec2x from correct volume statement
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Question 14

a) i)	$f(x) = x(x-2)^4$ Product rule $f'(x) = (x) \cdot 4(x-2)^3 + (x-2)^4$ $= (x-2)^3(4x + (x-2))$ $= (x-2)^3(5x-2)$	1 mark correct solution to 2 nd last line that shows factorising
ii)	If $(x-2)^3(5x-2) = 0$ $x = 2 \text{ or } x = \frac{2}{5}$ at $x = 2$ $y = 0$ At $x = 0$ $f'(0) = 16$ at $x = \frac{2}{5}$ $f'\left(\frac{2}{5}\right) = 0$ at $x = 1$ $f'(1) = -3$ at $x = 2$ $f'(2) = 0$ and at $x = 3$ $f'(3) = 3 \therefore$ if $x < 2$ $f'(x) < 0$ and if $x > 2$ $f'(x) > 0$ hence a minimum at $x = \frac{2}{5}$ $y = 2.6$ (1d.p.) if $x < \frac{2}{5}$ $f'(x) > 0$ and if $x > \frac{2}{5}$ $f'(x) < 0$ hence a maximum also at $x = 0$ $y = 0$	2 marks full correct solution 1 mark coordinates of both points or correct determination of both stationary points as max then min
iii)		2 marks correct solution or correct based on student answer 1 mark for either max or min graphed correct with correct shape and no further mistakes
b) i)	$y = f(x)$ is a reflection so is turned upside down and hence becomes $-\cos(2x)$ and because it is reflected in the line $y = 1$ it moves up 2 units hence + 2 $\therefore f(x) = -\cos(2x) + 2$ $= 2 - \cos(2x)$	2 marks correct solution 1 mark for either negative cos or +2

b ii)	$\int_0^{2\pi} 2 - \cos(2x) dx$ from the graph this is made up of 8 identical parts taken between 0 and $\frac{\pi}{2}$ taken below the curve and above the line $y = 1$ $\begin{aligned} \text{Area} &= 8 \int_0^{\frac{\pi}{2}} 2 - \cos(2x) - 1 dx \\ &= 8 \int_0^{\frac{\pi}{2}} 1 - \cos(2x) dx \\ &= 8 \left[x - \frac{1}{2} \sin(2x) \right]_0^{\frac{\pi}{2}} \\ &= 8 \left[\left(\frac{\pi}{2} - \frac{1}{2} \sin(\pi) \right) - 0 \right] \\ &= 8 \left[\frac{\pi}{2} - 0 \right] \\ &= 4\pi \text{ units}^2 \end{aligned}$ OR $\begin{aligned} &\int_0^{2\pi} 2 - \cos(2x) - \cos(2x) dx \\ &= \int_0^{2\pi} 2 - 2\cos(2x) dx \\ &= \left[2x - \sin(2x) \right]_0^{2\pi} \\ &= (2(2\pi) - \sin(4\pi)) - (0 - \sin(0)) \\ &= 4\pi \text{ units}^2 \end{aligned}$	2 marks correct solution 1 mark correct applicant of areas between curves and satisfactory integration and substitution
c) i)	In $\triangle AXM$ and $\triangle BXC$ $\angle MAX = \angle BCX$ (alternate angles on parallel lines, AD and CB opposite sides of rectangle $ABCD$ given) $\angle AXM = \angle BXC$ (vertically opposite $\angle$) $\therefore \triangle AXM \parallel \triangle BXC$ (equiangular)	2marks correct solution 1mark if did not give 2 correct reasons or no statement of test used for similarity

c ii)	$\frac{AM}{BC} = \frac{1}{2} \text{ (M is midpoint of AD (given))}$ and $AD = BC$ (opp sides of rectangle $ABCD$) $\therefore \frac{AX}{CX} = \frac{1}{2} \text{ (corresponding sides in similar } \Delta \text{)}$ $CX = 2AX$ now $AC = AX + CX$ $AC = AX + 2AX$ $AC = 3AX$ Show $3CX = 2AC$ $\begin{aligned} \text{LHS} &= 3CX \\ &= 3(2AX) \\ &= 6AX \\ \text{RHS} &= 2AC \\ &= 2(3AX) \\ &= 6AX \end{aligned}$ LHS=RHS as required Or Using similar Δ as above $\frac{AX}{BC} = \frac{1}{2}$ $2AX = CX \text{ and } AX = AC - CX$ $\therefore 2(AC - CX) = CX$ $2AC - 2CX = CX$ $2AC = 3CX$ $\therefore 3CX = 2AC$	1 mark correct solution Must logically show required statement
iii)	In ΔABC $\angle B$ is 90° $AC^2 = AB^2 + BC^2$ using $3CX = 2AC \therefore AC = \frac{3CX}{2}$ and $AB = 2AD$(given) and $BC = AD$ $\therefore BC = \frac{AB}{2}$ $\left(\frac{3CX}{2} \right)^2 = AB^2 + \left(\frac{AB}{2} \right)^2$ $\frac{9CX^2}{4} = AB^2 + \frac{AB^2}{4}$ $9CX^2 = 4AB^2 + AB^2$ $9CX^2 = 5AB^2$	2 marks correct solution 1mark correct substitution into Pythagoras or Correct use of part ii) and logical progress

Question 15

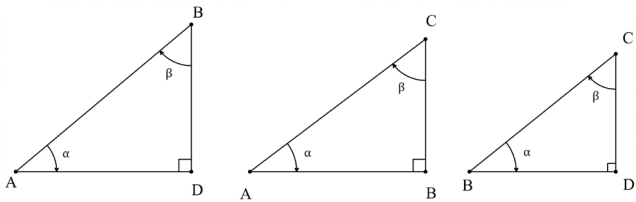
a	$\int_0^{\sqrt{e}} \frac{x^2}{x^3 + e} dx$ $= \frac{1}{3} \int_0^{\sqrt{e}} \frac{3x^2}{x^3 + e} dx$ $= \frac{1}{3} \left[\ln x^3 + e \right]_0^{\sqrt{e}}$ $= \frac{1}{3} (\ln (\sqrt{e})^3 + e - \ln 0^3 + e)$ $= \frac{1}{3} (\ln(e\sqrt{e} + e) - \ln e)$ $= \frac{1}{3} \ln\left(\frac{e\sqrt{e} + e}{e}\right)$ $= \frac{1}{3} \ln(\sqrt{e} + 1)$ $= \frac{\ln(\sqrt{e} + 1)}{3}$	3 marks 1st mark for the correct integral 2nd mark for a correct substitution 3rd mark for correct solution using log laws
bi	6% p.a. = 0.5% per month On 1st December, the \$4000 principal amount will have increased by 0.5% 11 times. $4000(1.005)^{11} = \$4225.58$	1 mark
bii	$A_0 = 4000$ $A_1 = 4000(1.005) + M$ Note: For most of January, there is only \$4000 in Jacqueline's account, so the bank rewards her with 0.5% interest on that amount (not \$(4000+M))\$. $A_2 = A_1(1.005) + M$ $= (4000(1.005) + M) \times 1.005 + M$ $= 4000(1.005)^2 + M(1.005) + M$ $= 4000(1.005)^2 + M(1 + 1.005)$ Keep going until a pattern can be seen. $A_3 = 4000(1.005)^3 + M(1 + 1.005 + 1.005^2)$ $A_n = 4000(1.005)^n + M(1 + 1.005 + 1.005^2 + \dots + 1.005^{n-1})$	3 marks correct solution (or correct given errors from part (a)) 2 marks for a minor error e.g. adding deposits before compounding interest or treating as 12 deposits (i.e. 31st Dec) 1 mark for some understanding, using geometric series

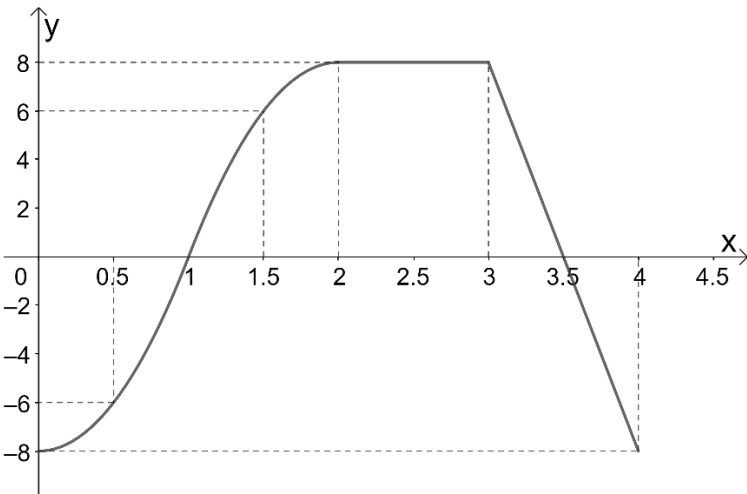
	On 1st December, there have been 11 deposits, and the value of the account is \$7000. $7000 = 4000(1.005)^{11} + M(1 + 1.005 + 1.005^2 + \dots + 1.005^{10})$ The sum of the geometric series is: $S_{11} = \frac{1.005^{11} - 1}{1.005 - 1}$ $M \times S_{11} = 7000 - 4000(1.005)^{11}$ $M = \frac{7000 - 4000(1.005)^{11}}{S_{11}}$ $= \$245.98$	
ci	C is one of three points of intersection of the cubic and the line. To find points of intersection: $4x^3 - 9x^2 = -5x$ $4x^3 - 9x^2 + 5x = 0$ $x(4x^2 - 9x + 5) = 0$ $x(4x - 5)(x - 1) = 0$ $x = 0, 1, \frac{5}{4}$ Note: From diagram, C is the middle point of intersection y-coord of C: $y = -5(1)$ $= -5$ $\therefore C(1, -5)$	2 marks 1 mark for solving the equations simultaneously Note: Several students mistook $x = \frac{5}{4}$ as the x-coordinate of C. Unfortunately this made part (iii) unprovable.
cii	$d = \frac{ ax_1 + by_1 + c }{\sqrt{a^2 + b^2}}$ $(x_1, y_1) = A(x, y)$ Equation of line OC in general form: $y = -5x \Rightarrow 5x + y = 0$	2 marks (no explanation required for removing the absolute values) 1 mark for a correct expression using perpendicular distance formula with $y = -5x$ and $A(x, y)$

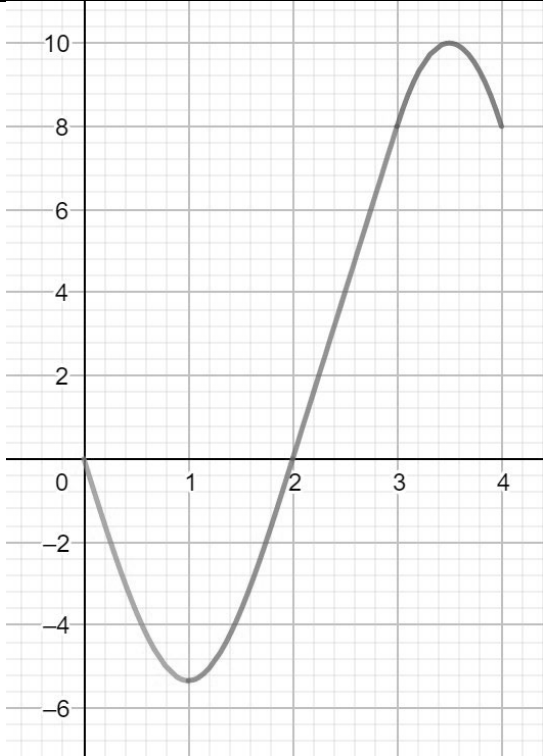
	$d = \frac{ 5(x) + 5(y) }{\sqrt{5^2 + 1^2}}$ $= \frac{ 5x + y }{\sqrt{26}}$ Since A lies on the cubic, the coordinates of A satisfy $y = 4x^3 - 9x^2$ $\therefore d = \frac{ 5x + 4x^3 - 9x^2 }{\sqrt{26}}$ $= \frac{ 4x^3 - 9x^2 + 5x }{\sqrt{26}}$ Since the cubic is above the line in the domain $0 < x < 1$, $4x^3 - 9x^2 > -5x$ $4x^3 - 9x^2 + 5x \text{ is positive}$ $\therefore d = \frac{4x^3 - 9x^2 + 5x}{\sqrt{26}}$ Note: What makes this question confusing for students is the non-numerical use of perpendicular distance formula, and the use of the pronumerals x and y as coordinates of point A, where x is between 0 and 1. It may be helpful to write x_A and y_A to denote the x and y-coordinates of point A.	There were several unsuccessful attempts to 'reverse-engineer' the solution from the result. The best method in this case is to simply use the perpendicular distance formula with point A and line OC to obtain an initial expression (1 mark), then try to obtain the result.
ciii	ΔAOC $A = \frac{1}{2}bh$ $= \frac{1}{2}(OC) \times d$ Pythagoras: $OC = \sqrt{1^2 + 5^2}$ $= \sqrt{26}$	1 mark

	$A = \frac{1}{2} \times \sqrt{26} \times \frac{1}{\sqrt{26}} (4x^3 - 9x^2 + 5x)$ $= \frac{1}{2} (4x^3 - 9x^2 + 5x)$ $= 2x^3 - \frac{9}{2}x^2 + \frac{5}{2}x$	
civ	$A = 2x^3 - \frac{9}{2}x^2 + \frac{5}{2}x$ $\frac{dA}{dx} = 6x^2 - 9x + \frac{5}{2}$ Stationary points: $\frac{dA}{dx} = 0$ $6x^2 - 9x + \frac{5}{2} = 0$ $12x^2 - 18x + 5 = 0$ $x = \frac{18 \pm \sqrt{(-18)^2 - 4(12)(5)}}{2(12)}$ $= \frac{18 \pm \sqrt{84}}{24}$ $= \frac{18 \pm 2\sqrt{21}}{24}$ $= \frac{9 \pm \sqrt{21}}{12}$ $= 0.36811... \text{ or } 1.131...$ The x-coord of A is between 0 and 1 Therefore $x = 0.36811$ must be the value that produces a maximum area $A = 2(0.36811)^3 - \frac{9}{2}(0.36811)^2 + \frac{5}{2}(0.36811)$ $= 0.41 \text{ square units (2dp)}$	3 marks 1st mark solving $A'=0$ 2nd mark for choosing the correct value of x 3rd mark for substituting into A Note: Students did well attempting part (iv), even if they weren't able to prove the previous parts. Note: Normally we are required to determine the nature of a stationary point, but in this case we are told already that point A is chosen so that the area is a maximum, and since $x=0.36811$ is the only stationary point between 0 and 1, we know it must be the maximum.

Question 16

a-i	 $\Delta ABD \parallel \Delta ABC$ $\therefore \frac{AB}{AC} = \frac{AD}{AB}$ $AB^2 = AC \cdot AD$	1 mark – showing both first two lines. Identifying the similar triangles and then final statement was not sufficient.
a-ii	$\Delta ADB \parallel \Delta BDC$ $\therefore \frac{BD}{CD} = \frac{AD}{BD}$ $BD^2 = CD \cdot AD$ $AB^2 + BD^2 = AC \cdot AD + CD \cdot AD$ $= AD(AC + CD)$ $= AD(AD + CD + CD)$ $= AD(AD + 2CD)$ Pythagorean solution also accepted.	2 marks – full solution 1 mark – showing derivation of $AB^2 + BD^2 = AC \cdot AD + CD \cdot AD$ Pythagorean method – stating of a number of triads was not sufficient unless progress made towards solution.
b-i	Decreasing therefore $\frac{dy}{dx} < 0$ true for $0 < x < 1 \text{ and } 3.5 < x < 4$ also accepted $x < 1$ or $x > 3.5$ (should not really be using $\geq$ or $\leq$)	1 mark

b-ii	 $\int_0^2 f(x) dx = 0$ – given $\int_2^3 f(x) dx = \text{Area of Rectangle} = 8$ $\int_3^{3.5} f(x) dx = -\int_{3.5}^4 f(x) dx$ $\therefore \int_3^4 f(x) dx = 0$ therefore $f(4) = 0 + 8 + 0 = 8$	1 mark – correct answer
b-iii	Max occurs at $x = 3.5$ At $x = 2$, particle returns to origin. Distance from origin to the maximum equals area of trapezium. $A = \frac{8}{2}(1 + 1.5) = 10$	1 mark – correct answer

b-iv		2 marks correct answer 1 mark – correct answer with $f(0)$ and $f(2)$ correct at same displacement and one of $f(3.5)$ or $f(4)$
c-i	$P = 520\,000$ $r = \frac{8.4}{12}\% = .7\% \Rightarrow R = 1.007$ $M =$ monthly repayment $A_1 = P \times R - M$ $A_2 = A_1 \times R - M = PR^2 - MR - M$ $A_3 = A_2 \times R - M = PR^3 - MR^2 - MR - M$ $A_3 = 520\,000 \times 1.007^3 - M(1.007^2 + 1.007 + 1)$ $= 520\,000 \times 1.007^3 - M \frac{1.007^3 - 1}{1.007 - 1}$	2 marks – derivation of correct formula – preference to have used the sum of geometric series 1 mark – correct formula without showing derivation.

c- ii	$n = 15 \times 12 = 180$ $A_{180} = 0$ $A_{180} = 520\,000 \times 1.007^{180} - M \frac{1.007^{180} - 1}{1.007 - 1}$ $M \frac{1.007^{180} - 1}{1.007 - 1} = 520\,000 \times 1.007^{180}$ $M = 520\,000 \times 1.007^{180} \times \frac{1.007 - 1}{(1.007^{180} - 1)}$ $M = 5090.2104$ $M = 5090.21$	2 marks - correct solution 1 mark – correct expression for $A_{180}=0$
c- ii	$A_{24} = 520\,000 \times 1.007^{24} - 5090.21 \frac{(1.007^{24} - 1)}{1.007 - 1}$ $= 482243.89$ Amount owing = $482243.89 - 20000 = 462243.89$ $An = 462243.89 \times 1.007^n - 5090.21 \frac{(1.007^n - 1)}{1.007 - 1}$ $0 = 462243.89 \times 1.007^n - 5090.21((1.007^n - 1)) \times \frac{1000}{7}$ $0 = \frac{7}{1000} \times 462243.89 \times 1.007^n - 5090.21((1.007^n - 1))$ $-5090.21 = \left(\frac{7}{1000} \times 462243.89 - 5090.21 \right) (1.007^n)$ $1.007^n = \frac{-5090.210}{\frac{7}{1000} \times 462243.89 - 5090.21} = 2.744784$ $n(\log 1.007) = \log 2.744784$ $n = \frac{\log 2.744784}{\log 1.007} = 144.74747 \cong 145$	3 marks – correct answer 2 marks – show single term with n 1 mark – new amount owing